

# Boing Network — Development Recommendations & Enhancement Vision

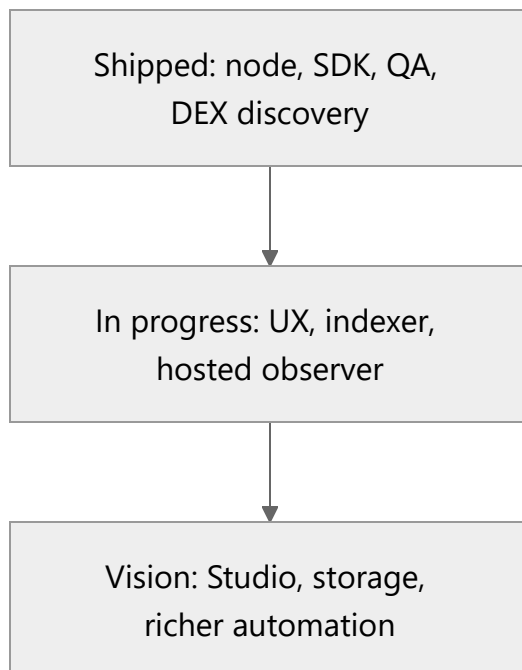
👤 **Everyday users:** this is a **vision** doc, not a user manual. Shipped products: [wallet](#), [explorer](#), [finance](#).

🔧 **Developers:** mix of implemented SDK/RPC work and **non-normative TBD**. Prefer [RPC-API-SPEC.md](#) and [boing-sdk](#) for what exists today.

👨‍💻 **Operators:** implementation priority matrix is in §1.5 — do not schedule TBD items as if they are committed.

**Purpose:** Strategic recommendations and vision for SDK, decentralized automation, ecosystem, intent-based execution, developer incentives, and Boing Studio.

**References:** [BUILD-ROADMAP.md](#), [BOING-BLOCKCHAIN-DESIGN-PLAN.md](#), [NETWORK-COST-ESTIMATE.md](#), [DECENTRALIZATION-AND-NETWORKING.md](#) — *Cryptographic verification for automation is [Appendix: Cryptographic verification](#) at the end of this document.*



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## Part 1: Development Recommendations

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**Goal:** Minimal friction for developers; first-class automation; robust ecosystem.

## 1.1 Boing SDK Refinement & Enhancement

### Expand Tooling & Developer Experience (DX)

| Recommendation                       | Description                                                                                                      | Priority |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------|----------|
| <b>IDE Extensions</b>                | VS Code / Cursor plugins: syntax highlighting for Boing contracts, code completion, debugging, direct deployment | Medium   |
| <b>CLI Auto-completion</b>           | Shell auto-completion for <code>boing</code> CLI (bash, zsh, fish)                                               | High     |
| <b>Code Snippets &amp; Templates</b> | Library of contract patterns, dApp templates, automation recipes via <code>boing init</code>                     | High     |

### Multi-Language Support

| Recommendation                   | Description                                                                            | Priority |
|----------------------------------|----------------------------------------------------------------------------------------|----------|
| <b>TypeScript/JavaScript SDK</b> | Official bindings for web frontends; JSON-RPC client, tx signing helpers               | High     |
| <b>Python SDK</b>                | Scripting, data analysis, bot tooling                                                  | Medium   |
| <b>Rust SDK</b>                  | Core; ensure <code>boing-sdk</code> crate exposes clean APIs for contracts and clients | High     |

### Documentation & Tutorials

| Recommendation               | Description                                                | Priority |
|------------------------------|------------------------------------------------------------|----------|
| <b>Step-by-step guides</b>   | First dApp, first automation, first cross-chain flow       | High     |
| <b>API reference</b>         | Auto-generated from code; detailed RPC, SDK, contract APIs | High     |
| <b>Interactive tutorials</b> | Learn-by-doing playgrounds                                 | Medium   |
| <b>Example dApps</b>         | Reference implementations (DeFi, NFT, automation)          | High     |

## Error Handling & Debugging

| Recommendation            | Description                                    | Priority |
|---------------------------|------------------------------------------------|----------|
| Meaningful error messages | SDK returns actionable, human-readable errors  | High     |
| Network diagnostics       | RPC health, chain height, sync status, latency | Medium   |
| Transaction tracing       | Debug failed txs, gas usage breakdown          | Medium   |

## 1.2 Decentralized Automation Features

### Executor Incentives & Slashing

| Recommendation               | Description                                                                                          | Priority |
|------------------------------|------------------------------------------------------------------------------------------------------|----------|
| <b>Reward model</b>          | Define incentive mechanism for Decentralized Executors: rewards for correctness and timely execution | High     |
| <b>Penalty mechanism</b>     | Slashing or reputation penalties for missed tasks, incorrect execution, malicious behavior           | High     |
| <b>Staking for executors</b> | Executors stake BOING; slashed on failure; rewards distributed per successful execution              | High     |

### Advanced Scheduling & Triggers

| Recommendation              | Description                                                                      | Priority |
|-----------------------------|----------------------------------------------------------------------------------|----------|
| <b>Complex cron</b>         | Beyond basic cron: "first Monday of month," "every 3h between 9–17"              | Medium   |
| <b>Conditional triggers</b> | "When on-chain event X, run Y" — predicates on contract state, oracle data, time | High     |
| <b>Event-driven hooks</b>   | Listen to block events, tx receipts, state changes                               | High     |

### User-Facing Automation

| Recommendation                        | Description                                                                 | Priority |
|---------------------------------------|-----------------------------------------------------------------------------|----------|
| <b>Visual workflow builder</b>        | No-code/low-code UI for Zap-style workflows; abstract blockchain complexity | Medium   |
| <b>Domain-specific language (DSL)</b> | Readable DSL for automation rules; compiles to on-chain logic               | Medium   |
| <b>Automation templates</b>           | Pre-built: recurring transfer, DCA, limit orders, cross-chain swap          | High     |

## Security & Verifiability

| Recommendation         | Description                                                                         | Priority | Status                                                              |
|------------------------|-------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------|
| Execution verification | Cryptographic proof of correct execution; ZK or optimistic for off-chain automation | Medium   | ✓ See <a href="#">appendix below</a>                                |
| Executor attestation   | Executors sign execution reports; slashing for fraud                                | High     | ✓ <code>ExecutorAttestation</code> in <code>boing-automation</code> |
| Access control         | Granular permissions: who can trigger, modify, cancel tasks                         | High     | —                                                                   |
| Gas abstraction        | Meta-txs, gas sponsorship for user-facing automation                                | High     | —                                                                   |

## 1.3 Overall Network Enhancements

### Success-Based dApp Incentives

| Recommendation               | Description                                                            | Priority |
|------------------------------|------------------------------------------------------------------------|----------|
| <b>Value cap per dApp</b>    | Epoch or monthly cap (e.g. max 10M BOING/month per dApp owner)         | High     |
| <b>Governance parameter</b>  | Cap and $f(\text{metrics})$ formula adjustable via on-chain governance | High     |
| <b>Success metrics</b>       | Transaction count, fees, volume, unique users, TVL                     | High     |
| <b>Transparent reporting</b> | Dashboard / SDK for dApp owners to track earned incentives             | Medium   |
| <b>Automated payout</b>      | Distribution contract; formula-driven payouts                          | High     |

### Cross-Chain Interoperability

| Recommendation                        | Description                                                                        | Priority |
|---------------------------------------|------------------------------------------------------------------------------------|----------|
| <b>Multi-network product surfaces</b> | Detail how Boing SDK and automation work across boing.finance and bridged networks | High     |
| <b>Cross-chain SDK helpers</b>        | Asset transfers, remote contract calls, event listening across chains              | High     |
| <b>Bridge standards</b>               | IBC-style or custom; trust-minimized design                                        | Medium   |

### Security & Scalability

| Recommendation             | Description                                                                     | Priority |
|----------------------------|---------------------------------------------------------------------------------|----------|
| <b>Security audits</b>     | Independent audit of consensus, execution, automation contracts                 | High     |
| <b>Formal verification</b> | Critical components (consensus, VM core)                                        | Medium   |
| <b>Scalability roadmap</b> | Clear path from dev → private testnet → mainnet; throughput and latency targets | High     |

Community & Ecosystem

| Recommendation     | Description                               | Priority |
|--------------------|-------------------------------------------|----------|
| Grant programs     | Attract early dApp developers             | Medium   |
| Hackathons         | Focus on SDK and decentralized automation | Medium   |
| Community channels | Discord, forums for support and feedback  | Medium   |

Economic Model Transparency

| Recommendation | Description                                                             | Priority |
|----------------|-------------------------------------------------------------------------|----------|
| Documentation  | Tokenomics, validator incentives, developer treasury clearly documented | High     |
| Transparency   | Public dashboards for emissions, fees, incentive distributions          | Medium   |



## 1.4 Advanced Decentralization & P2P

**Goal:** Robust, censorship-resistant peer discovery and networking. See [DECENTRALIZATION-AND-NETWORKING.md](#).

| Recommendation                  | Description                                                               | Priority |
|---------------------------------|---------------------------------------------------------------------------|----------|
| <b>DHT + gossip-first</b>       | Kademlia DHT with gossip overlay; minimize reliance on fixed bootnodes    | High     |
| <b>Bootnode rotation</b>        | Governance-rotated or community-funded bootnodes; no single choke point   | High     |
| <b>Sybil/eclipse resistance</b> | Reputation or PoS within DHT; diversify connections; re-verify peer lists | High     |
| <b>WebRTC/WebSockets</b>        | Browser-based light clients; decentralized signaling                      | Medium   |
| <b>Incentivized relayers</b>    | Reward relay nodes (Filecoin-style) for robust relay network              | Medium   |
| <b>VDF/VRF</b>                  | Verifiable randomness for leader selection; fair ordering                 | High     |
| <b>Trustless bridges</b>        | ZKP or MPC relayers with slashing; avoid federated multisig               | High     |

## 1.5 Implementation Priority Matrix

| Area                   | Immediate (0–4 weeks)                                                                                         | Short-term (1–3 months)                                    | Medium-term (3–6 months)                                |
|------------------------|---------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|---------------------------------------------------------|
| <b>SDK</b>             | <code>boing init</code> , <code>boing dev</code> , <code>boing deploy</code> ; CLI auto-completion; templates | TS/JS client; IDE extension; interactive tutorials         | Python SDK; visual workflow builder; Boing Studio       |
| <b>Automation</b>      | Native scheduler; basic triggers; executor staking                                                            | Conditional triggers; gas abstraction; DSL design          | Visual builder; execution verification                  |
| <b>dApp Incentives</b> | Value cap spec; <code>f(metrics)</code> formula; governance param                                             | Incentive contract; payout distribution; dynamic royalties | Dashboard; transparent reporting                        |
| <b>Cross-chain</b>     | RPC/API docs; Boing-specific flows                                                                            | Cross-chain helpers; intent signing format                 | Bridge standards; trustless bridges; meta-router        |
| <b>Security</b>        | Internal review; test coverage                                                                                | External audit planning; continuous audit cadence          | Formal verification                                     |
| <b>P2P</b>             | libp2p swarm; basic gossip                                                                                    | DHT; gossip-first; bootnode rotation                       | Sybil/eclipse resistance; WebRTC; incentivized relayers |

## Part 2: Enhancement Vision

**Goal:** Amplify authenticity, uniqueness, and practical implementation.

## 2.1 Elevating Cross-Chain DeFi Coordination

### Intent-Based Transaction Execution

| Aspect          | Description                                                                                                                               |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Concept         | Users declare <i>intent</i> (e.g., "swap X for Y across any chain at best price") rather than specifying a rigid path.                    |
| Role of Boing   | Decentralized automation layer identifies, orchestrates, and executes the optimal cross-chain path. Aggregates liquidity; MEV protection. |
| Differentiation | Boing as indispensable "meta-router" for DeFi; abstracts chain-specific complexity.                                                       |

### Native Cross-Chain Liquidity Provisioning

| Aspect    | Description                                                                                    |
|-----------|------------------------------------------------------------------------------------------------|
| Mechanism | Protocol-level support for liquidity providers to offer capital to cross-chain pools on Boing. |
| Benefit   | Earn fees from orchestrated swaps across connected chains; Boing as central liquidity hub.     |

## 2.2 Advanced Protocol-Native Developer Incentives

### Dynamic Fee Allocation & Developer Royalties

| Aspect    | Description                                                                                                                      |
|-----------|----------------------------------------------------------------------------------------------------------------------------------|
| Concept   | dApps specify royalty splits from user fees. Example: 1% to original contract developer, 0.5% to library, rest to dApp treasury. |
| Value Cap | Per-dApp owner cap (see <a href="#">NETWORK-COST-ESTIMATE.md</a> ) remains governance-controlled.                                |
| Benefit   | Self-sustaining ecosystem of composable dApps; incentives flow to contributors.                                                  |

### Reputation-Based Resource Access

| Aspect  | Description                                                                                                                                  |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Concept | Soulbound contribution credentials; high-reputation developers receive priority tx processing, discounted gas, or increased API rate limits. |
| Benefit | Incentivizes high-quality contributions; rewards long-term builders.                                                                         |

## 2.3 Decentralized Storage Integration

### Native Permanent Archival

| Aspect             | Description                                                                                                       |
|--------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>Integration</b> | Filecoin, Arweave, or similar for permanent, tamper-proof archival of historical chain state, tx logs, dApp data. |
| <b>Benefit</b>     | Enhanced data availability; censorship resistance; audit trails for DeFi.                                         |

### Decentralized Content Delivery (CDN)

| Aspect           | Description                                                            |
|------------------|------------------------------------------------------------------------|
| <b>Mechanism</b> | SDK tools to deploy dApp frontends and static assets to IPFS/Filecoin. |
| <b>Benefit</b>   | Frontends as censorship-resistant as backends.                         |

## 2.4 Authentic Developer Experience

### Boing SDK with Built-in AI Assistance

| Aspect          | Description                                                                                               |
|-----------------|-----------------------------------------------------------------------------------------------------------|
| Capabilities    | Code generation for common patterns; automatic contract vulnerability scanning; optimization suggestions. |
| Differentiation | Exceptionally developer-friendly; leverages AI to reduce friction.                                        |

### Boing Studio — Integrated Development Environment

| Aspect   | Description                                                                                                  |
|----------|--------------------------------------------------------------------------------------------------------------|
| Concept  | Web-based or local IDE (Remix-style for Boing): SDK integration, templates, debugging, one-click deployment. |
| Features | Syntax highlighting, deployment to testnet/mainnet, debugging tools, tailored for Boing's unique features.   |

## 2.5 Practical Implementation Focus

### Security Audits

| Approach          | Description                                                                                                        |
|-------------------|--------------------------------------------------------------------------------------------------------------------|
| <b>Continuous</b> | Rigorous, independent audits of core protocol, custom VM, consensus, and automation — not one-time before mainnet. |
| <b>Scope</b>      | Consensus, execution, decentralized automation contracts, bridge logic.                                            |

### Community Engagement

| Approach          | Description                                                           |
|-------------------|-----------------------------------------------------------------------|
| <b>Validators</b> | Clear docs; easy node operation; educational programs.                |
| <b>Developers</b> | Grant programs; hackathons; SDK-focused events.                       |
| <b>Users</b>      | Community channels; transparent governance; participation incentives. |

## Network-Wide Enhancements: Implemented & Planned

This section tracks **overall network** enhancements (not limited to one pillar). See also [BOING-NETWORK-ESSENTIALS.md](#) for the six pillars and [QUALITY-ASSURANCE-NETWORK.md](#) for the QA pillar.

## Implemented (recent pass)

| Area             | Enhancement                                               | Where                            |
|------------------|-----------------------------------------------------------|----------------------------------|
| Priorities       | Sixth pillar (True QA) in README                          | README.md                        |
| Block production | Re-insert txs into mempool on execution/consensus failure | mempool.reinsert, block_producer |
| Mempool          | drain_for_block returns Vec for re-insert                 | mempool.rs                       |
| Roadmap          | Protocol QA phase (4.6)                                   | BUILD-ROADMAP.md §4.6            |
| RPC spec         | boing_qaCheck, QA error codes -32050, -32051              | RPC-API-SPEC.md                  |
| Essentials doc   | Six pillars, tech stack, key docs                         | BOING-NETWORK-ESSENTIALS.md      |
| RUNBOOK          | Six pillars section, getBalance/getAccount in RPC table   | RUNBOOK.md                       |



## Enhancements by area (status)

| Area                              | Done                                                                       | Planned / reference                                                                                                                                 |
|-----------------------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Resilience</b>                 | Re-insert txs on failure                                                   | P2P connection management, formal verification (BUILD-ROADMAP §4.3)                                                                                 |
| <b>Protocol QA</b>                | boing-qa crate stub, RPC spec                                              | Node integration, community pool, governance of rules (BUILD-ROADMAP §4.6)                                                                          |
| <b>Developer experience</b>       | Official <code>boing-sdk</code> (TypeScript), tutorial scripts, RPC probes | CLI auto-completion; IDE extensions; guides and API reference depth (this doc §1.1; BUILD-ROADMAP §5.5; <a href="#">NEXT-STEPS-FUTURE-WORK.md</a> ) |
| <b>Decentralization &amp; P2P</b> | —                                                                          | DHT, bootnode rotation, peer scoring (BUILD-ROADMAP §2.2)                                                                                           |
| <b>Automation &amp; UX</b>        | —                                                                          | Conditional triggers, automation SDK (this doc §1.2; BUILD-ROADMAP §5.7)                                                                            |
| <b>Security</b>                   | —                                                                          | Bug bounty, post-quantum path, audit (BUILD-ROADMAP §4.3)                                                                                           |

## Priority overview

- **High impact next:** boing-qa node integration maturity; CLI auto-completion; `boing-sdk` depth (RPC integration CI, indexer helpers — [NEXT-STEPS-FUTURE-WORK.md](#)); DHT and peer scoring.
- **Ongoing:** [BUILD-ROADMAP.md](#) and this document for phased tasks; near-term backlog routing in [NEXT-STEPS-FUTURE-WORK.md](#).

## Implementation Status

| Area                                        | Implemented | Notes                                                                                      |
|---------------------------------------------|-------------|--------------------------------------------------------------------------------------------|
| <b>boing init</b>                           | ✓           | Scaffolds Cargo.toml, README, boing.json, src/lib.rs                                       |
| <b>boing dev</b>                            | ✓           | Spawns boing-node via cargo                                                                |
| <b>boing deploy</b>                         | ✓           | Connects to RPC, validates reachability                                                    |
| <b>boing metrics register</b>               | Stub        | CLI accepts params; backend TBD                                                            |
| <b>CronSchedule / Scheduler</b>             | ✓           | <code>boing-automation</code> crate                                                        |
| <b>Trigger / TriggerCondition</b>           | ✓           | Block height, balance, timestamp                                                           |
| <b>ExecutorIncentive</b>                    | ✓           | Design: reward, slash, min stake                                                           |
| <b>ExecutorAttestation</b>                  | ✓           | Signed execution reports; verify()                                                         |
| <b>ExecutionProof, ZkpProof, FraudProof</b> | ✓           | Verification types (ZKP/FraudProof placeholders)                                           |
| <b>OracleAttestation</b>                    | ✓           | Oracle data + quorum signatures                                                            |
| <b>dApp incentive formula</b>               | ✓           | <code>DappMetrics</code> , <code>dapp_incentive()</code> , <code>VALUE_CAP_PER_DAPP</code> |

## Cross-References

- **NEXT-STEPS-FUTURE-WORK.md** — Consolidated engineering backlog: where to file infra / CI upgrades, indexer and native AMM follow-ups, and optional small PR slices alongside this vision doc
- **BOING-NETWORK-ESSENTIALS.md** — Six pillars, design philosophy, priorities, tech stack
- **QUALITY-ASSURANCE-NETWORK.md** — Protocol QA (sixth pillar); automation and community pool

- **BUILD-ROADMAP.md** — Phase 5.5 (Developer Experience), 5.6 (Success-Based dApp Incentives), 5.7 (Decentralized Automation)
- **Appendix (this doc)** — Cryptographic verification for on-chain and off-chain automation
- **NETWORK-COST-ESTIMATE.md** — Phased Cost Overview; economic parameters
- **BOING-BLOCKCHAIN-DESIGN-PLAN.md** — Innovation sections; UX & Human-Centered; Technical Innovations
- **DECENTRALIZATION-AND-NETWORKING.md** — Advanced peer discovery; DHT; gossip-first; WebRTC; relayers
- **SECURITY-STANDARDS.md** — Protocol, network, application, operational security

## Appendix: Cryptographic verification for decentralized automation

**Purpose:** Define how automated tasks are cryptographically verified for trustless, reliable execution.

**References:** [BUILD-ROADMAP.md](#), [BOING-BLOCKCHAIN-DESIGN-PLAN.md](#)

### Overview

The Boing Network's native decentralized automation layer requires robust cryptographic verification to ensure trustless execution for developers, validators, and users. The verification method depends on whether the task is **on-chain** or involves **off-chain** elements.

### 1. On-Chain Automation

For automation that resides entirely within the Boing blockchain (scheduled smart contract calls, native cron, auto-compounding), verification is inherent:

#### 1.1 Cryptographically Signed Transactions

- Every automated action generates a **signed transaction**.
- Transactions are signed by the Decentralized Executor or protocol-level automation contract.
- All network nodes **mathematically verify** digital signatures; invalid signatures cause rejection.
- **Implementation:** Uses Ed25519 via [SignedTransaction](#) ; verification in consensus and execution layers.

#### 1.2 Deterministic Execution

- Smart contract code is **deterministic**.
- Given the same starting state and inputs, every honest node produces the same final state.
- Any deviation indicates a fault.
- **Implementation:** Boing VM (interpreter, bytecode spec) is deterministic; parallel execution preserves determinism via access-list batching.

#### 1.3 Consensus Mechanism

- PoS + BFT finality ensures a **supermajority** of validators agree on block order and validity.
- Once a block is finalized, execution is **immutable** and verified by the network.
- **Implementation:** HotStuff consensus;  $2f+1$  quorum; equivocation detection and slashing.

## 2. Off-Chain Automation & External Data

When tasks require off-chain computation or external data (e.g. "When token X hits price Y, execute Z"), additional cryptographic techniques apply:

### 2.1 Zero-Knowledge Proofs (ZKPs)

| Aspect                   | Description                                                                                                                                                                 |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Mechanism</b>         | An Executor proves a computation was performed correctly <i>without revealing inputs or computation details</i> .                                                           |
| <b>Boing Application</b> | dApps perform complex calculations off-chain (e.g. risk assessment), submit a concise ZKP to the chain, and an on-chain contract validates it before triggering automation. |
| <b>Status</b>            | Design target; SDK to provide ZKP generation helpers.                                                                                                                       |

### 2.2 Optimistic Rollups / Fraud Proofs

| Aspect                   | Description                                                                                                                                                                                                 |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Mechanism</b>         | Results are optimistically assumed correct. During a challenge period, anyone can submit a <b>Fraud Proof</b> (cryptographic evidence of incorrect execution). If fraud is proven, the Executor is slashed. |
| <b>Boing Application</b> | Cost-effective for tasks where a challenge delay is acceptable; scalable off-chain verification.                                                                                                            |
| <b>Status</b>            | <code>FraudProof</code> type in automation crate; integration TBD.                                                                                                                                          |

### 2.3 Decentralized Oracle Networks

| Aspect                   | Description                                                                                                             |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Mechanism</b>         | Multiple oracle nodes aggregate external data, cryptographically sign it, and provide attestations.                     |
| <b>Boing Application</b> | Native automation integrates oracle data so conditions like "token X price = Y" are verifiable via oracle attestations. |
| <b>Status</b>            | <code>OracleAttestation</code> design; oracle network integration TBD.                                                  |

## 2.4 Attestations by Decentralized Executors

| Aspect                   | Description                                                                                                                                          |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Mechanism</b>         | Executors cryptographically sign execution reports. Stake provides economic incentive for honesty; slashing punishes incorrect or malicious signing. |
| <b>Boing Application</b> | Executors sign messages confirming action and parameters; protocol verifies signatures and applies slashing.                                         |
| <b>Status</b>            | <code>ExecutorAttestation</code> implemented; signed execution reports.                                                                              |

## 3. Protocol-Level Integration

### 3.1 Native Scheduler & Trigger Verification

- Scheduler and trigger components **demand and verify** appropriate proofs or attestations per task type.
- On-chain tasks: standard tx verification.
- Off-chain tasks: ZKP, Fraud Proof, or Executor Attestation.

### 3.2 Transparent Incentives & Slashing

- **Rewards:** Executors earn BOING for correct, timely execution.
- **Slashing:** Incorrect execution, missed tasks, or fraud proofs trigger stake slashing.
- **Implementation:** `ExecutorIncentive`, `ExecutorRegistration`; slashing wired to consensus equivocation and automation verification outcomes.

### 3.3 Boing SDK Support

- SDK provides simplified interfaces for:
  - ZKP generation for off-chain dApp logic.
  - Interaction with the native oracle layer.
  - Submitting and verifying Executor attestations.

## Summary: Verification by Task Type

| Task Type                      | Verification Method                             |
|--------------------------------|-------------------------------------------------|
| On-chain scheduled call        | Signed tx + deterministic execution + consensus |
| On-chain cron/trigger          | Same as above                                   |
| Off-chain compute (ZKP)        | ZKP validated on-chain                          |
| Off-chain compute (optimistic) | Fraud Proof during challenge period             |
| External data condition        | Oracle attestations                             |
| Executor-initiated action      | Executor attestation (signed report) + slashing |

## Implementation

| Type                             | Location                                    | Description                                                         |
|----------------------------------|---------------------------------------------|---------------------------------------------------------------------|
| <code>ExecutorAttestation</code> | <code>boing-automation::verification</code> | Signed execution report; <code>new()</code> , <code>verify()</code> |
| <code>ExecutionProof</code>      | <code>boing-automation::verification</code> | Enum: Attestation, Zkp, FraudProof                                  |
| <code>ZkpProof</code>            | <code>boing-automation::verification</code> | Placeholder for ZKP bytes                                           |
| <code>FraudProof</code>          | <code>boing-automation::verification</code> | Evidence of incorrect execution                                     |
| <code>OracleAttestation</code>   | <code>boing-automation::verification</code> | Oracle data + quorum signatures                                     |

See [boing-automation](#) crate.

*Boing Network — Authentic. Decentralized. Optimal. Sustainable.*